LEVEL 18 — Unique and Set Operations in NumPy

Line-by-line easy examples with outputs.

1. np.unique()

Removes duplicate values and returns unique values.

Code:

```
import numpy as np

arr = np.array([10, 20, 10, 30, 20, 40])
result = np.unique(arr)
print(result)
```

Output: [10 20 30 40]

2. np.intersect1d()

Returns values common to both arrays.

Code:

```
a = np.array([10, 20, 30, 40])
b = np.array([30, 40, 50, 60])
result = np.intersect1d(a, b)
print(result)
```

Output: [30 40]

3. np.union1d()

Combines both arrays and removes duplicates.

Code:

```
a = np.array([10, 20, 30])
b = np.array([30, 40, 50])
result = np.union1d(a, b)
print(result)
```

Output: [10 20 30 40 50]

4. np.setdiff1d()

Returns values present in the first array but not the second.

Code:

```
a = np.array([10, 20, 30, 40])
b = np.array([30, 40, 50])
result = np.setdiff1d(a, b)
print(result)
```

Output: [10 20]

5. np.setxor1d()

Returns values present in either array, but not in both.

Code:

```
a = np.array([10, 20, 30])
b = np.array([30, 40, 50])
result = np.setxor1d(a, b)
print(result)
```

Output: [10 20 40 50]

6. np.in1d() / Membership

Checks whether every value of the first array exists in the second. np.in1d() is deprecated in newer NumPy versions; use np.isin().

Code:

```
a = np.array([10, 20, 30, 40])
b = np.array([20, 40])
result = np.in1d(a, b)
print(result)
```

Output: [False True False True]

7. np.isin()

Modern membership check. Returns True when a value exists in the second array.

Code:

```
a = np.array([10, 20, 30, 40])
b = np.array([20, 40])
result = np.isin(a, b)
print(result)
```

Output: [False True False True]

8. Easy Difference

Intersection → common values.
Union → all unique values.
Difference → values in first array, not second.
Symmetric difference → values in either array, but not both.
Membership → checks whether values exist in another array.

9. Quick Revision

• np.unique() → removes duplicates.
• np.intersect1d() → common values.
• np.union1d() → all unique values.
• np.setdiff1d() → first minus second.
• np.setxor1d() → values in either, but not both.
• np.in1d() → older membership function.
• np.isin() → preferred modern membership function.